

Year 12 Trial Examination – Mathematics Extension 1

Question One	12 marks	(Start on a new page)	Marks
---------------------	-----------------	------------------------------	--------------

- | | | |
|----|---|----------|
| a) | If P is the point $(-3, 5)$ and Q is the point $(1, -2)$, find the co-ordinates of the point R which divides the interval PQ externally in the ratio $3:2$. | 2 |
| b) | When $(x + 3)(x - 2) + 2$ is divided by $x - k$, the remainder is k^2 . Find the value of k . | 2 |
| c) | Solve $\frac{x}{x-3} > 1$ | 3 |
| d) | Find the general solution of $\sin \theta = \cos \theta$ in radians | 2 |
| e) | Find the exact value of $\int_0^{\frac{\pi}{2}} 2 \sin^2 x \, dx$ | 3 |

Question Two	12 marks	(Start on a new page)	
---------------------	-----------------	------------------------------	--

- | | | |
|-------|--|----------|
| a) | Draw a graph of a function $y = f(x)$ for $1 \leq x \leq 2$ such that $\frac{dy}{dx} > 0$ and $\frac{d^2 y}{dx^2} < 0$ for $1 \leq x \leq 2$ | 2 |
| b) | Differentiate: | |
| i) | $\frac{1}{1 + 4x^2}$ | 2 |
| ii) | $e^{2x} \log_e 2x$ | 2 |
| c) i) | Show that $x^2 + 4x + 13 = (x + 2)^2 + 9$ | 1 |
| ii) | Hence find $\int \frac{1 \, dx}{x^2 + 4x + 13}$ | 2 |
| d) | Evaluate $\int_0^{\sqrt{3}} x \sqrt{x^2 + 1} \, dx$ using the substitution that $u = x^2 + 1$ | 3 |

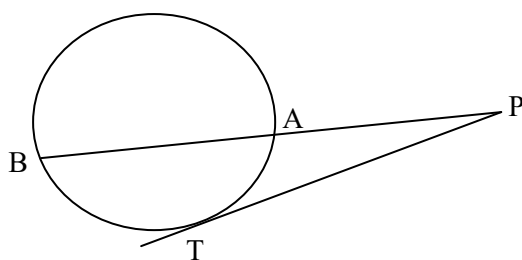
Question Three **12 marks** **(Start on a new page)** **Marks**

a) Given $f(x) = \frac{x-1}{x+2}$

i) Write an expression for the inverse function $f^{-1}(x)$ 1

ii) Write down the domain and range of $f^{-1}(x)$ 2

b)



PT is a tangent to circle ABT .
 PAB is a secant intersecting the circle in A and B .
 $PA = 8$ cm and $AB = 10$ cm.
 Find the length of PT giving reasons.

2

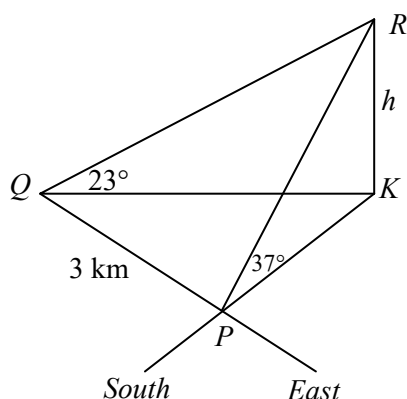
c) Find the gradients of the 2 lines which make angles of 45° with the line whose equation is $2x - 3y + 6 = 0$. 3

d) i) Express $\cos \theta + \sqrt{3} \sin \theta$ in the form $r \cos(\theta - \alpha)$ where $r > 0$ and $0 < \alpha < \frac{\pi}{2}$ 2

ii) Hence solve $\cos \theta + \sqrt{3} \sin \theta = 1$ for $-2\pi \leq \theta \leq 2\pi$ 2

Question Four **12 marks** **(Start on a new page)**

a)



The angle of elevation of a hill top from a place P due south of it is 37° .
 The angle of elevation of this same hill top from a place Q , due west of P , is 23° . The distance of Q from P is 3 km. If the height of the hill is h km.

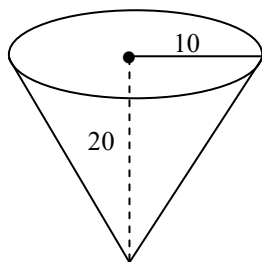
i) Prove that $PK = h \cot 37^\circ$ 2

ii) Find a similar expression for QK 2

iii) Hence, or otherwise calculate the height of the hill to two decimal places. 2

Question Four continued**Marks**

b)

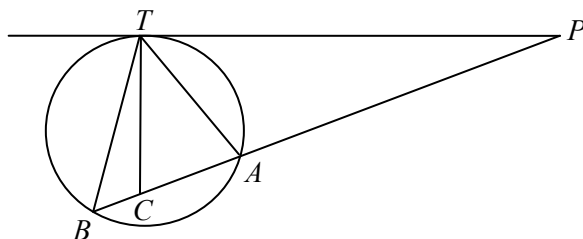


Water is running out of a filled conical funnel at the rate of $5 \text{ cm}^3 \text{ s}^{-1}$. The radius of the funnel is 10 cm and the height is 20 cm:

- i) How fast is the water level dropping when the water is 10 cm deep?
(answer in exact form) 4
- ii) How long does it take for the water to drop to 10 cm deep?
(answer to 2 decimal places) 2

Question Five **12 marks** (Start on a new page)

a)



PT is a tangent and PAB is a secant. $TC = TA$.

Prove $\angle BTC = \angle TPA$

3b) Given θ is acute:

- i) Write $\sin \frac{\theta}{2}$ in terms of $\cos \theta$ 1

- ii) Prove that $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$ 2

- iii) If $\sin \theta = \frac{4}{5}$, find the value of $\tan \frac{\theta}{2}$ 3

- c) Find $\frac{d}{dx} \cos^{-1}(\sin x)$ 3

Question Six	12 marks	(Start on a new page)	Marks
---------------------	-----------------	------------------------------	--------------

- a) i) Show that the sum of the cubes of three consecutive integers $(n-1), n, (n+1)$ is $3n^3 + 6n$. 2
- ii) Using part i), prove by mathematical induction, for all positive integers $n, n \geq 1$ that the sum of the cubes of the three consecutive integers is divisible by 9. 3
- b) Consider the variable point $P(x, y)$ on the parabola $x^2 = 2y$. The x value of P is given by $x = t$:
- i) write its y value in terms of t 1
- ii) write an expression, in terms of t , for the square of the distance, m from P to the point $(6, 0)$ 1
- iii) hence find the co-ordinates of P such that P is closest to the point $(6, 0)$. 5

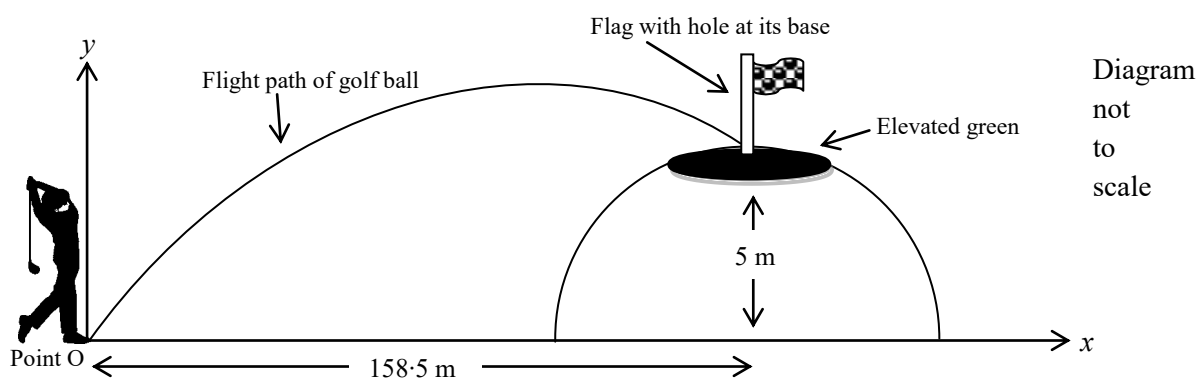
Question Seven	12 marks	(Start on a new page)	
-----------------------	-----------------	------------------------------	--

- a) Use the binomial expansion of $(1+x)^{2n}$ to show that:
- i) $1 - 2\binom{2n}{1} + 4\binom{2n}{2} - \dots + 4^n \binom{2n}{2n} = 1$ 2
- ii) $\binom{2n}{1} - 4\binom{2n}{2} + 12\binom{2n}{3} - \dots - n4^n \binom{2n}{2n} = -2n$ 2

Question Seven continued

Marks

- b) Mr Mac hits a golf ball from a point O towards a flat, elevated green as shown in the diagram below. The hole at the base of the flag is situated in the centre of the green:



The golf ball is projected from the point O with initial velocity of $v \text{ ms}^{-1}$ at an angle of α to the horizontal. You may assume the only force acting on the golf ball in flight is gravity which is approximately 10 ms^{-2} :

- i) Taking the point O as the origin, show the parametric equations of the flight path of the golf ball are given by $x = vt \cos \alpha$ and $y = -5t^2 + vt \sin \alpha$ 2
- ii) If the angle of projection is given by $\alpha = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$ find v which will enable Mr Mac to hit the ball directly into the hole on the green. The hole is situated 158.5 m horizontally and 5 m vertically from point O. Answer to the nearest integer. 3
- iii) Using your value of v from part (ii), at what angle does the ball strike the green? (answer to the nearest degree) 3

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

QUESTION 1

a) $P(-3, 5) \quad G(1, -2)$

$$x = \frac{-6-3}{-3+2} = 9$$

$$y = \frac{10+6}{-3+2} = -16$$

$$\therefore R(9, -16)$$

b) $P(k) = (k+3)(k+2) + 2 = k^2$

$$k-4=0$$

$$k=4$$

c) $\frac{x}{x-3} > 1 \quad , x \neq 3$

$$x(x-3) > (x-3)^2$$

$$x(x-3) - (x-3)^2 > 0$$

$$(x-3)(3) > 0$$

$$x > 3$$

d) $\sin \theta = \cos \theta$

$$\tan \theta = 1$$

$$\theta = \tan^{-1} 1 + n\pi$$

$$\theta = \frac{\pi}{4} + n\pi$$

e) $\int_0^{\frac{\pi}{2}} 2 \sin^2 x \, dx$

$$\int_0^{\frac{\pi}{2}} (1 - \cos 2x) \, dx$$

$$= \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} - \frac{1}{2} \sin \pi - 0$$

$$= \pi$$

QUESTION 2

$$2bi) \quad y = \frac{1}{1+4x^2}$$

$$y' = \frac{-8}{(1+4x^2)}$$

$$bii) \quad y = e^{2x} \log_e 2x$$

$$y = e^{2x} \left(2 \ln x + \frac{1}{x} \right)$$

$$ci) \quad \text{RHS} = (x+2)^2 + 9$$

$$= x^2 + 4x + 4 + 9$$

$$= x^2 + 4x + 13$$

$$= \text{LHS}$$

$$cii) \quad \int \frac{1dx}{x^2 + 4x + 13}$$

$$\int \frac{1dx}{(x+2)^2 + 9}$$

$$= \frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right) + c$$

$$2d) \quad \int_0^{\sqrt{3}} x\sqrt{x^2+1} \, dx$$

$$= \frac{1}{2} \int_1^4 \sqrt{u} \, du$$

$$= \frac{1}{3} \left[u^{\frac{3}{2}} \right]_1^4$$

$$= \frac{7}{3}$$

$$u = x^2 + 1, \quad du = 2x dx$$

$$x = \sqrt{3}, \quad u = 4$$

$$x = 0, \quad u = 1$$

Question 3

$$ai) \quad y = \frac{x-1}{x+2}$$

Question 3

$$\text{ai) } y = \frac{x-1}{x+2}$$

$$x = \frac{y-1}{y+2}$$

$$xy + 2x = y - 1$$

$$y(1-x) = 2x+1$$

$$y = \frac{2x+1}{1-x}$$

$$f^{-1} = \frac{2x+1}{1-x}$$

ii) Domain all real, $x \neq 1$

range all real, $y \neq -2$

$$b) PT^2 = BP \times PA \quad (\text{prod of secant and tangent})$$

$$= 18 \times 8$$

$$PT = 12$$

$$c) \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$1 = \frac{\frac{2}{3} - m_2}{1 + \frac{2}{3} m_2}$$

$$1 + \frac{2}{3} m_2 = \frac{2}{3} - m_2$$

$$\frac{1}{3} = \frac{5}{3} m_2$$

$$\therefore m_2 = \frac{-1}{5}$$

since $45^\circ + 45^\circ = 90^\circ$,

$$m_1 = 5$$

$$\begin{aligned} di) \cos \theta + \sqrt{3} \sin \theta &= r \cos(\theta - \alpha) \\ &= r \cos \theta \cos \alpha - r \sin \theta \sin \alpha \end{aligned}$$

$$r \cos \alpha = 1$$

$$r \sin \alpha = \sqrt{3}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \cos \theta + \sqrt{3} \sin \theta = 2 \cos\left(\theta - \frac{\pi}{3}\right)$$

$$ii) \cos \theta + \sqrt{3} \sin \theta = 1$$

$$2 \cos\left(\theta - \frac{\pi}{3}\right) = 1$$

$$\cos\left(\theta - \frac{\pi}{3}\right) = \frac{1}{2}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{-\pi}{3}, \frac{-5\pi}{3}, \frac{-7\pi}{3}$$

$$\theta = \frac{2\pi}{3}, \frac{-4\pi}{3}, 0, 2\pi, -2\pi$$

QUESTION 4

$$ai) \tan 37 = \frac{PK}{h}$$

$$PK = h \cot 37$$

$$a ii) \tan 23 = \frac{h}{QK}$$

$$QK = h \cot 23$$

$$4a iii) 3^2 + h^2 \cot^2 37 = h^2 \cot^2 23$$

$$h(\cot^2 23 - \cot^2 37) = 9$$

$$h = \frac{9}{(\cot^2 23 - \cot^2 37)}$$

$$h = 1.54 \text{ km}$$

$$\begin{aligned}
 4bi) \quad v &= \frac{1}{3} \pi r^2 h \\
 &= \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h, \quad r = \frac{h}{2} \\
 &= \frac{1}{12} \pi h^3
 \end{aligned}$$

$$\begin{aligned}
 \frac{dv}{dt} &= \frac{1}{4} \pi h^2 \\
 \frac{dh}{dt} &= \frac{dh}{dv} \times \frac{dv}{dt} \\
 &= \frac{4}{\pi h^2} \times -5 \\
 &= \frac{-20}{\pi h^2}
 \end{aligned}$$

$$\text{when } h = 10, \quad \frac{dh}{dt} = \frac{-1}{5\pi} \text{ cm / s}$$

$$ii) \quad \frac{dv}{dt} = -5$$

$$v = -5t + c$$

$$\text{when } t = 0, v = \frac{1}{12} \pi (20)^3 = \frac{250\pi}{3}$$

$$\frac{250\pi}{3} = -5t + \frac{2000\pi}{3}$$

$$5t = \frac{1750\pi}{3}$$

$$t = \frac{350\pi}{3}$$

$$t = 367s$$

Question 5

5a) $\angle PTA = \angle BTA$ (angle between tan and chord)

$\angle TPA + \angle PTA = \angle TAC$ (ext angle of Δ)

$\therefore \angle TCA = \angle TPA + \angle PTA$ (equal angles of isos Δ)

$\angle BTC = \angle TCB - \angle BTA$ (equal angles of isos Δ)

$\therefore \angle BTC = \angle TPA$

$$bi) \cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}$$

$$2 \sin^2 \theta = 1 - \cos \theta$$

$$\sin \theta = \sqrt{\frac{1 - \cos \theta}{2}}$$

$$bii) RHS = \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 + \left(2 \cos^2 \frac{\theta}{2} - 1 \right)}$$

$$= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\left(2 \cos^2 \frac{\theta}{2} \right)}$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}$$

$$= \tan \frac{\theta}{2} = RHS$$

$$\begin{aligned}
 \text{biii) } \tan \frac{\theta}{2} &= \frac{\sin \theta}{1 + \cos \theta} \\
 &= \frac{\frac{4}{5}}{\left(1 + \frac{3}{5}\right)} \\
 &= \frac{1}{2}
 \end{aligned}$$

$$c) \frac{d}{dx} \cos^{-1}(\sin x)$$

$$= \frac{-\frac{d}{dx} \sin x}{\sqrt{1 - (\sin x)^2}}$$

$$= \frac{-\cos x}{\sqrt{\cos^2 x}}$$

$$= -1 \text{ for 1st and 4th quad}$$

$$= 1 \text{ for 2nd and 3rd quad, } \cos x \neq 0$$

Question 6

$$\begin{aligned}
 ai) S_{(n)} &= (n-1)^2 + n^3 + (n+1)^3 \\
 &= n^3 - 3n^2 + 3n - 1 + n^3 + n^3 + 3n^2 + 3n + 1 \\
 &= 3n^3 + 6n \\
 &= RHS
 \end{aligned}$$

$$ii) S_{(1)} = 9, \therefore S_{(1)} \text{ is divisible by } 9$$

Assume that $S_{(n)}$ is divisible by 9 when $n = k$

ie $S_{(k)} = 9m$, then

$$\begin{aligned}
 S_{(k+1)} &= 3(k+1)^3 + 6(k+1) \\
 &= 3k^3 + 6k + 9(k^2 + k + 1) \\
 &= 9m + 9(k^2 + k + 1)
 \end{aligned}$$

$\therefore S_{(k+1)}$ is divisible by 9. Hence by the pple of MI, the statement is true for all $n \geq 1$

$$b) x^2 = 2y$$

$$i) x = t, t^2 = 2y$$

$$\therefore y = \frac{t^2}{2}$$

$$\begin{aligned}
 ii) m^2 &= (t-6)^2 + \left(\frac{t^2}{2}\right)^2 \\
 &= t^2 - 12t + 36 + \frac{t^4}{4}
 \end{aligned}$$

$$iii) \frac{dm^2}{dt} = 2t - 12 + t^3 = 0$$

$$(t-2)(t^2 + 2t + 6) = 0$$

$$t = 2, t^2 + 2t + 6 = 0$$

$$P(2, 2) \quad \Delta = 4 - 24 = -20$$

$\therefore$ no soln

QUESTION 7

$$ai) \quad (1+x)^{2n} = {}^{2n}C_0 x^0 + {}^{2n}C_1 x^1 + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n}$$

$$\text{let } x = -2$$

$$\therefore (-1)^{2n} = 1 + {}^{2n}C_1 (-2) + {}^{2n}C_2 (-2)^2 + \dots + {}^{2n}C_{2n} (-2)^{2n}$$

$$\therefore [(-1)^2]^n = 1 - 2 {}^{2n}C_1 + 4 {}^{2n}C_2 + \dots + [(-2)^2]^n {}^{2n}C_{2n}$$

$$\text{ie. } 1 = 1 - 2 {}^{2n}C_1 + 4 {}^{2n}C_2 + \dots + 4^n {}^{2n}C_{2n} \quad \text{as reqd}$$

aii) Differentiate wrt x

$$2n(1+x)^{2n-1} = 0 + {}^{2n}C_1 + 2x {}^{2n}C_2 + 3x^2 {}^{2n}C_3 + \dots + 2nx^{2n-1} {}^{2n}C_{2n}$$

$$\text{now let } x = -2$$

$$2n(-1)^{2n-1} = {}^{2n}C_1 - 4 {}^{2n}C_2 + 12 {}^{2n}C_3 + \dots + 2n {}^{2n}C_{2n} [(-2)^2]^n \cdot (-2)^{-1}$$

$$2n(-1) = {}^{2n}C_1 - 4 {}^{2n}C_2 + 12 {}^{2n}C_3 + \dots + n {}^{2n}C_{2n} [4]^n \cdot (-1)$$

$$\therefore -2n = {}^{2n}C_1 - 4 {}^{2n}C_2 + 12 {}^{2n}C_3 + \dots - n {}^{2n}C_{2n} [4]^n \quad \text{as reqd}$$

Question 7b continued

7bi) horizontal $\ddot{x} = 0$

$$\dot{x} = c_1$$

$$\dot{x} = v \cos \alpha$$

$$x = \int v \cos \alpha dt$$

$$= vt \cos \alpha + c_2$$

$$t = 0, x = 0 \therefore c_2 = 0$$

$$\therefore x = vt \cos \alpha$$

vertical $\ddot{y} = -10$

$$\dot{y} = -10t + c_3$$

$$\dot{y} = v \sin \alpha$$

$$t = 0,$$

$$\therefore c_3 = v \sin \alpha$$

$$\dot{y} = -10t + v \sin \alpha$$

$$y = \int -10t + v \sin \alpha dt$$

$$\therefore y = -5t^2 + vt \sin \alpha$$

$$ii) \alpha = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$\cos \alpha = \frac{\sqrt{3}}{2} \quad \sin \alpha = \frac{1}{2}$$

$$vt \cos \alpha = x$$

$$t = \frac{x}{v \cos \alpha}$$

sub into

$$y = -5t^2 + vt \sin \alpha$$

$$= -5 \left(\frac{x^2}{v^2 \cos^2 \alpha} \right) + \frac{vx}{v \cos \alpha} \times \sin \alpha$$

$$= \frac{-5x^2 \sec^2 \alpha}{v^2} + x \tan \alpha$$

$$= \frac{-5x^2}{v^2} (1 + \tan^2 \alpha) + x \tan \alpha$$

$$\text{using } x = 158.5, y = 5, \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\text{then } 5 = \frac{-5 \times 158.5^2 \left(1 + \frac{1}{3} \right)}{v^2} + 158.5 \times \frac{1}{\sqrt{3}}$$

$$v^2 = \frac{5 \times 158.5^2 \times 4 \times \sqrt{3}}{(158.5 \times 3 - 15\sqrt{3})}$$

$$v^2 = 1935.98$$

$$v = 43.999 \text{ m/s}$$

$$v = 44 \text{ m/s}$$

$$\begin{aligned} 7biii) \quad y &= \frac{-5x^2}{v^2 \cos^2 \alpha} + x \tan \alpha \\ &= \frac{-5x^2}{44^2 \left(\frac{\sqrt{3}}{2} \right)^2} + x \left(\frac{1}{\sqrt{3}} \right) \\ &= \frac{-5}{1452} x^2 + \frac{\sqrt{3}}{3} x \end{aligned}$$

$$\dot{y} = \frac{-5}{726} x + \frac{\sqrt{3}}{3}$$

$$m_{\tan} = \frac{-5 \times \frac{317}{2}}{726} + \frac{\sqrt{3}}{3}$$

$$\text{acute } \tan \theta = 27^\circ$$

$\therefore$ ball strikes at 153°

End of assessment